

j
CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter row and column of matrix 1:");
        int row1=sc.nextInt();
        int col1=sc.nextInt();
        System.out.println("Enter row and column of matrix 2:");
        int row2=sc.nextInt();
        int col2=sc.nextInt();
        int[][] mat1=new int[row1][col1];
        int[][] mat2=new int[row2][col2];
        int[][] product=new int[row1][col2];
        System.out.println("Enter elements of mat1:");
        for(int i=0;i<row1;i++){
            for(int j=0;j<col1;j++){
                mat1[i][j]=sc.nextInt();
            }
        }
        System.out.println("Enter elements of mat2:");
        for(int i=0;i<row2;i++){
            for(int j=0;j<col2;j++){
                mat2[i][j]=sc.nextInt();
            }
        }
        for(int i=0;i<row1;i++){
            for(int j=0;j<col2;j++){
                for(int k=0;k<col1;k++){
                    product[i][j]+=mat1[i][k]*mat2[k][j];
                }
            }
        }
        System.out.println("Mat Sum=");
        for(int i=0;i<row1;i++){
            for(int j=0;j<col2;j++){
                System.out.print(product[i][j]+" ");
            }
            System.out.println();
        }
        sc.close();
    }
}
```

INPUT

2
2
2
2
1
2
5
3
2
3
4
1

OUTPUT

Enter row and column of matrix 1:
Enter row and column of matrix 2:
Enter elements of mat1:
Enter elements of mat2:
Mat Sum=
10 5
22 18

j
CODE

```
import java.util.Scanner;
public class PerfectNumber{
    public PerfectNumber(int n){
        System.out.println("First "+n+" Perfect numbers are:");
        int count=0;
        int num=2;
        while(count<n){
            int sum=0;
            for(int i=1;i<num;i++){
                if(num%i==0){
                    sum+=i;
                }
            }
            if(sum==num){
                count++;
                System.out.print(num+(count==n?"":" "),");
            }
            num++;
        }
        System.out.println();
    }
    public static void main(String[] args){
        Scanner sc= new Scanner(System.in);
        System.out.println("N=");
        int n=sc.nextInt();
        new PerfectNumber(n);
    }
}
```

INPUT

3

OUTPUT

N=
First 3 Perfect numbers are:
6,28,496

j
CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the income:");
        double income=sc.nextDouble();
        double tax=0;
        if(income<=150000){
            tax=0;
        }else if(income>=150001 && income<=300000){
            tax=income*0.10;
        }else if(income>=300001 && income<=500000){
            tax=income*0.20;
        }else if(income>=500001){
            tax=income*0.30;
        }
        System.out.println("Tax="+tax);
    }
}
```

INPUT

200000

OUTPUT

Enter the income:
Tax=20000.0

J

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int uppercount=0;
        int lowercount=0;
        int numcount=0;
        System.out.println("Enter * to exit...");
        while(true){
            System.out.println("Enter any character:");
            String input=sc.next();
            char ch=input.charAt(0);
            if(ch=='*'){
                break;
            }
            if(Character.isUpperCase(ch)){
                uppercount++;
            }else if(Character.isLowerCase(ch)){
                lowercount++;
            }else if(Character.isDigit(ch)){
                numcount++;
            }
        }
        System.out.println("Total count of lower case:"+lowercount);
        System.out.println("Total count of upper case:"+uppercount);
        System.out.println("Total count of numbers =" +numcount);
    }
}
```

INPUT

W
d
A
G
g
H
*

OUTPUT

Enter * to exit...
Enter any character:
Enter any character:
Enter any character:
Enter any character:
Enter any character:
Enter any character:
Enter any character:
Total count of lower case:2
Total count of upper case:4
Total count of numbers =0